

AI3D Co-Create with AI3D Render

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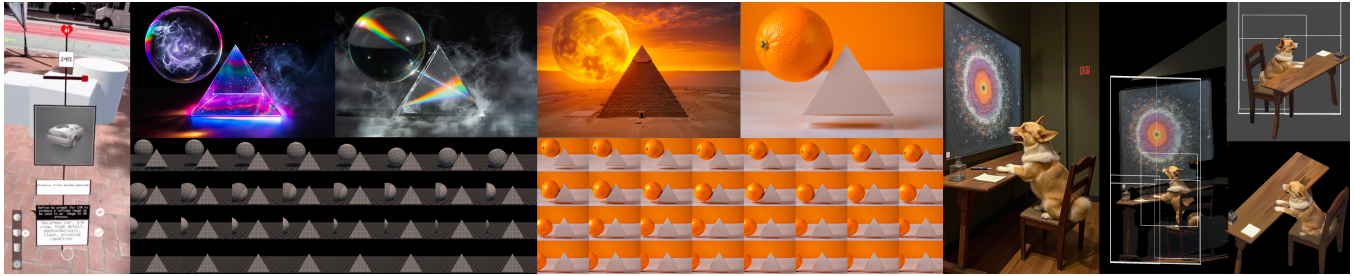


Figure 1: 1) LCM real-time maps a rough AI3D primitive scene into the desired 3D object; 2) Top: first frames; bottom left: AI3D rough video defining the motion; bottom right: generated result from the rightmost top frame; 3) Bounding box (x, y, label) detection with MoGe depth estimation determines the z coordinate, followed by flux-kontext in-painting (image-to-mesh).

Abstract

AI3D Co-Create is an app and framework supporting both realtime and "rendered" 3D multimodal controlled inputs in creating gen AI content with a large context length creative AI assistant (image to image to 3D and also using 3D to control gen AI video). By prompting directly inside of the spatial user interface, and getting 3D results back immediately, the user can maintain flow state while creating and manipulating 3D content. In cases where we have to await final video render time as in AI3D Render, we maintain flow by allowing for the motion to be defined in realtime by the user. The app is designed to be cross-platform (iOS/Android/VisionOS/Mac/Win) to experiment with intuitive 3D creation HCI across all relevant interaction modalities (traditional AR/XR mobile/headset and hand tracking + glasses-free 3D such as the Sony Spatial Reality Display). We show our solution to be a new SOTA HCI paradigm for human + AI co-creation. (As a timely comparison, we give HCI control benchmarks for what our framework can do, which cannot be achieved through text or single frame image means through Veo 3, Flux Kontext, etc.)

CCS Concepts

• Human-centered computing → Human computer interaction (HCI); Mixed / augmented reality; Collaborative interaction; • Computing methodologies → Artificial intelligence.

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Keywords

Gen AI, HCI, Realtime, Hybrid Methods for 3D and video creation

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1 Introduction

1.1 Co-Creation: real-time mode.

Co-Creation sends each interaction and operation to a large context length LLM - which "grows" to remember your artistic tendencies and is either multi-modal or works with a VLM to co-create an intelligent feedback flow to help you create your vision faster.

1.2 Scene Composition

1.2.1 (Entire) Image to 3D Worlds - Worldmatica Mode. Beyond simple image to 3D, we can piecewise de-compose any image into N objects for N image to 3D operations to create entire worlds.

Geometry Estimation for precise z axis placement in detected objects from image. We can use an object detector to detect known objects in an image, and send each bounding box (x, y, label) to an image to 3D model to generate 3D models of each identified object in the image. We can use boolean labels and flux-kontext to remove objects to send a coherent image to 3D input in (or, object-segment and inverse-object-inpaint). For example, "remove object; inpaint occluded parts" Using MoGe, we can extract the z position to place each 3D model with xyz accuracy to recreate the entire scene.

Skybox to 3D worlds. We extend the method into a $N \times M$ grid of a skybox, to fully extract and create the 3D objects of an entire world.

(We can also use video in-painting and clever stitching methods in AniSkyboxAI to create looping generative video from a skybox image.)

1.2.2 Robust Realtime Co-Creation Methods.

AI3D Primitives. Using a realtime img2img model such as LCM we are able to in realtime turn a "easel-based screenshot" of a rough arrangement of primitives + text prompt into the image of a complex 3D model, for a subsequent image to 3D operation to generate a new geometry. [Chang 2024b]

AI3D Sculpt. One can also use the same method above to live sculpt into the final desired 3D model faster. Each sculpt action prompts the easel to generate a screenshot of the potential desired final output, until the user applies the final image to 3D operation. [Chang 2024d] AI3D Desktop Skeuomorphic Interfaces Other skeuomorphic interfaces beyond the easel-based method may be found in the author's previous apps AI3D Desktop [Chang 2024c] and AI3D Camera Obscura [Chang 2024a] and also paper sketch to 3D models and beyond Napkinmatic [Chang 2023a] [Chang 2023b] [Chang 2024e] and DrawmaticAR [Chang 2020].

1.2.3 Generative Placements.

Object Replacement from Image. We can also extract rough shapes from an image, whether from identifying bounding boxes and putting each to a fast lo-fi image2mesh model such as SF3D [et al 2024], to put through the process described above.

MCP. Our system is compatible with MCP [Anthropic 2024] servers to allow for Claude or another similar LLM to place primitives and other objects to create scenes, for subsequent img2img2mesh.

1.3 AI3D Render: Long async gen AI operations become the new "render time" bottlenecks

For many use-cases of 3D creation, the end result is the rendered video, and thus we present a method called "AI3D Render", where rather than traditional rendering, we are sending specific control data to a gen AI video model to produce an end result that we would normally have to "render".

Using the Go-with-the-Flow approach on CogVidX [et al 2025a], we guide video diffusion models by warping Gaussian noise with optical flow extracted from a rough version of the target motion—typically over the first 49 frames (variable depending on the model). This structured noise enables the model to generate videos that follow the desired motion while preserving high per-frame quality.

First Frame + gaussian noise motion. We can define the entire look using the first frame, and utilize noise-only motions for the rest. For example, we can use a controlnet-depth model that turns the first frame of a primitives-based animation into our desired scene; and we utilize the motion from the rest to "render" the entire segment. Utilizing a classical CG animation analogy: This can be likened to a mesh (first frame) having separate animation (gaussian noise motion), where different meshes with similar rigs (different first frames with the same controlnet-depth structure) can be animated with the same animation data. This allows us to add animations that require the third animation, such as spinning; this cannot be done expressively with current SOTA text2video models.

Precision Walk Path through an Image. Using MoGe [et al 2025b], an (Entire) Image to 3D method above, we can create a sparse 3D scene from a single image, where we can effectively "enter the world of a painting", similar to [Chang 2015]. We can then use the motion from animating a virtual camera to AI3D render a video walking through the scene. Text prompting an exact path is not possible with the latest SOTA Gen AI text to video models.

"Waypoint-Guided 3D Camera Motion in Generative Video". We extend motion control by introducing "waypoint-based camera navigation," where a virtual camera follows a predefined 3D path through a reconstructed scene. Starting from a single image, we estimate an implicit 3D mesh or depth structure using monocular depth prediction or ControlNet-depth conditioning. This structure serves as a scaffold for planning camera trajectories—e.g., dolly-ins, orbits, or even AI-driven pathfinding through the scene.

These trajectories are converted into "optical flow fields," which are then used to warp Gaussian noise via the Go-with-the-Flow approach [et al 2025a]. The result is a coherent video where the generated frames respect both the spatial structure of the scene and the intended camera motion. This framework not only enables cinematic controls such as "virtual cinematography," but also opens the door to "autonomous agents navigating within generative environments," blending principles of video synthesis with classical AI navigation and 3D graphics.

Using Image2Video2Video to Generate Accurate Spatial Data for Gaussian Splats (Worldmatica approach). Extending the virtual camera to a rotating camera, we can effectively cue the model into generating additional accurate spatial data for Gaussian Splats, for us to, in full circle, re-create the entire (partially generated) original 3D scene from a single image.

2 Conclusion

The above showcases a level of expressivity that cannot be achieved from traditional tools or the models themselves.

We hope these examples of realtime trans-reality connecting human expressiveness with gen AI models helps inspire more work with novel HCI designed especially for humans to be able to "break out of the textbox/console" in interacting with AI [Chang 2024f].

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